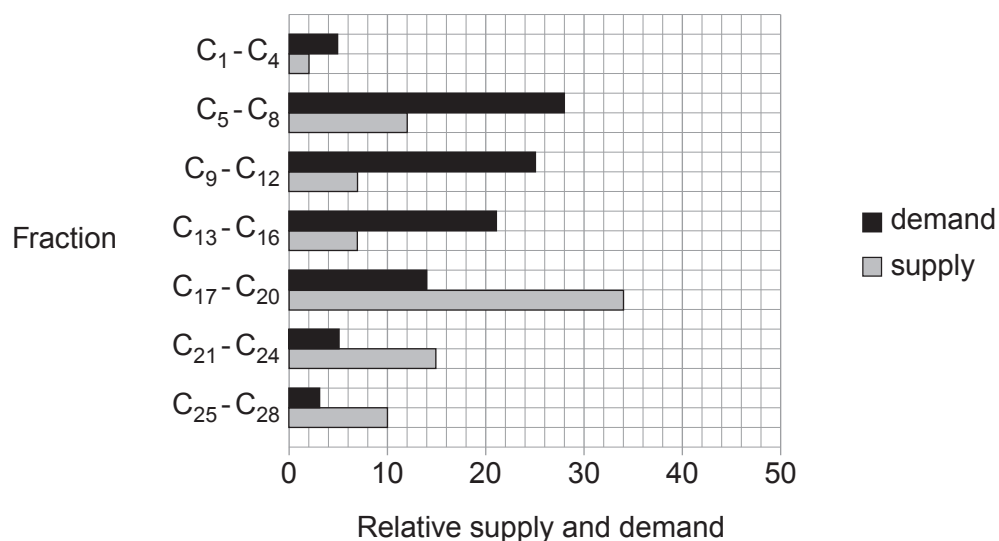


5. (a) The bar chart shows the relative supply and demand for some fractions obtained from crude oil.



- (i) Use the chart to describe how the **difference** between supply and demand of the fractions changes as the chain length increases. [2]

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- (ii) The process of cracking is used to overcome the problem of insufficient supply of some fractions.

I. State what is meant by *cracking* and give the conditions needed for the process. [2]

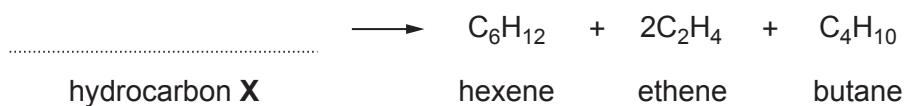
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II. When hydrocarbon **X** is cracked, it forms hexene, ethene and butane.

Give the molecular formula of hydrocarbon **X** to complete the equation. [1]



- (b) C_4H_{10} has two isomers.

(i) Give the meaning of the term *isomers*. [1]

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(ii) Draw the structures of both isomers of C_4H_{10} . [2]

